

Investigating the Challenges Faced by Women on Software Engineering: a Grey Literature Study

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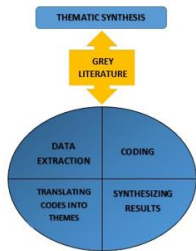
Slide 1

Research Goal

- Women face numerous challenges, including pay disparities, harassment, lack of professional recognition, and exclusion from leadership roles, despite having equivalent qualifications to their men counterparts [1, 2, 3, 4].
- Goal:
 - To identify and analyze the challenges faced by women in the professional technology sector, based on everyday descriptions found in grey literature.

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Study Settings



Agenda

- Research Goal and Questions
- Study Settings
- Results and Discussions
- Highlights
- Future Work

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Research Questions

- RQ.1 What are the challenges that women face in the software industry?** This question aims to identify the challenges faced by women when working on a software project.
- RQ.2 What effects do challenges have on women in the software industry?** In this question, we seek to better understand how women feel when they face challenges in their daily lives in the software industry.
- RQ.3 How to build a more inclusive environment for women ICT professionals?** The purpose of this question is to identify possible solutions suggested by women so that the software industry is more accessible to them.

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Pró inserir essa figura pois trata da nossa metodologia de pesquisa, a forma como foi levantados os códigos e categorias. Segue o texto que trata da explicação:

The methodology applied contains the steps illustrated in this figure, which uses thematic synthesis divided into 4 stages: data extraction, coding, transformation of codes into categories and synthesis of results, which were applied to the grey

literature. Initially, we carried out a pilot analysis, where three articles were randomly selected to calibrate the researchers' perspectives. Two researchers read these articles separately with the aim of answering the following questions: "What are the challenges?", "What are the effects?" and "What are the possible solutions?" The extracted data were stored in a spreadsheet. The data were composed of recorded units, that is, text segments that answer the research questions. Finally, the researchers compared and discussed their findings in a meeting. This comparison confirmed that the researchers were aligned on the data collection approach to be used. Next, the same researchers read the remaining articles. They then held another meeting to compare their results and reached a consensus, resulting in three subsets of recorded units, each corresponding to one of the questions. Finally, we conducted a coding process to identify the main idea in a recorded unit [26]. The same researchers jointly coded all recorded units, and another researcher reviewed the extracted codes. We standardized the codes, looking for codes that had the same meaning. For example, we standardized the codes "male dominance," "mostly men," and "few women in the field" into the code "male dominance." In the end, we had the unified codes and their number of occurrences in each subset. We identified that many of the codes were interconnected, allowing us to group them into related categories. The first author organized the codes into these categories, while the last author reviewed and refined the groupings. For example, the codes "stereotypes," "microaggressions," and "toxic masculinity" were grouped under the overarching category of "Stereotypes and Negative Attitudes." This approach to categorizing codes was applied consistently across all subsets.

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Study Settings

- We used a [set of articles posted in the Dev Community](#).
- We identified 85 articles.
- We selected a set of **44 articles** suitable for analysis.
- We carried out a [thematic synthesis](#) considering the 44 articles.
- We performed a [coding process](#).
- We identified many codes.

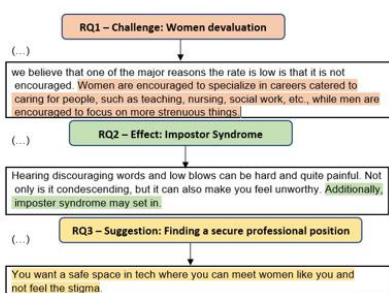


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These articles were published between early 2018 and the second half of 2023. The highest publication frequency was in 2020, with six articles, while the lowest was in 2018, with only one article. We also identified that the authors were mainly young women from various locations, performing different roles in the software industry, including developers (back-end and front-end), systems analysts, engineers, and testers.

Study Settings



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Esse slide exemplifica como foi feita a identificação dos códigos apartir dos artigos ligados, por exemplo o trecho da **RQ1 – Challenge: Women devaluation** (Desafio: Desvalorização das mulheres) – Destaca que as mulheres são encorajadas a seguir carreiras tradicionalmente associadas a cuidar de pessoas (como ensino, enfermagem, assistência social), enquanto os homens são estimulados a buscar trabalhos mais exigentes fisicamente e mentalmente.

Cada cor dessa imagem representa um exemplo de cada uma das RQ's.

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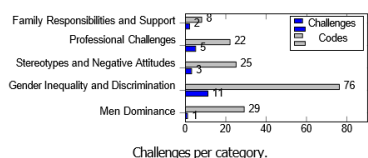


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RQ1: What are women's challenges in the software industry?

- We found **22 challenges** faced by women in the software industry and we organized them into **five categories**.



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Pró nesse slide é impossível identificar na imagem que a categoria **Gender Inequality and Discrimination** aparentemente é a de maior código, porem o que acontece ai é q a união de todos os códigos que formam essa categoria gerou essa quantidade(76) porem nenhum deles é maior que o código Men Dominance, o qual sozinho contem 29 aparições. E que é uma categoria de apenas 1 codigo.

Escrevo isso aqui pois alguém pode perguntar após a apresentação. Logo mais abaixo as definições de cada categoria resumidamente:

- The category **Men Dominance** is associated with the challenges that deals with men hegemony, with the code Men Dominance being the one in this category.
- **Stereotypes and Negative Attitudes** category refers to three challenges: stereotypes, microaggressions, and toxic masculinity, which comprises the way women are treated in the work environment analyzed in this study.
- The third category called **Gender Inequality and Discrimination** encompasses eleven challenges that are associated with the differences and pre-judgments that women suffer within the ICT area.
- **Professional Challenges** category covers five challenges faced by women that impede their development at work.
- The **Responsibilities and Family Support** category is associated with the duties and obligations that parents have in relation to the care, well-being and development of their children. Here, in the context of this study, these responsibilities are associated to a greater extent with women and also with the absence of emotional, practical or financial support from family members.

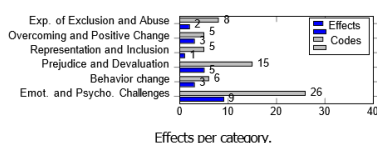
RQ1: What are women's challenges in the software industry?

- The category **Men Dominance** is associated with the challenges that deals with men hegemony, with the code *Men Dominance* being the one in this category. This code also have the highest number of occurrences (28). It means that the gender disparity might influence how women feel and behave.

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RQ2: What effects do challenges have on women in the software industry?

- In total, we identified **23 effects** resulting from the challenges mentioned in the previous slide, distributed across **six categories**.



Effects per category.

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- The category of **Emotional and Psychological Effects** is associated with the effects that deal with the emotional and psychological adversities faced by software engineers.
- The **Behavior Change** category is associated with the different stances that women need to take in an attempt to gain recognition in the technology sectors.
- The **Prejudice and Devaluation** category refers to the way women are treated in the workplace.
- The **Representation and Inclusion** category addresses the lack of women in the software industry, particularly in leadership environments.
- The **Overcoming and Positive Change** category it is characterized by processes that involve the ability to face challenges and adversities in a constructive way, resulting in personal and professional growth.

- The category **Experience of Exclusion and Abuse** portrays factors that keep women away from the technology field.

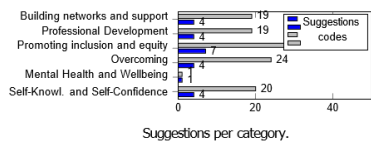
RQ2: What effects do challenges have on women in the software industry?

- The category of **Emotional and Psychological Effects** is associated with the effects that deal with the emotional and psychological adversities faced by software engineers. *Imposter Syndrome* is the most cited effect and was reported in the article (A1), as follows: "Hearing discouraging words and low blows can be difficult and quite painful. Not only is it condescending, but it can also make you feel unworthy."

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RQ3: How can we create a more inclusive environment for ICT professionals?

- In total, we found **24 suggestions**. They were grouped into **six categories**.



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- **Self-Knowledge and Self-Confidence** category is related to suggestions involved with aspects of women's personal security in themselves.
- Other category is **Mental Health and Wellbeing** that deals with women mental health care and had only one occurrence, despite this being a frequently cited problem, both as a challenge and as an effect.
- The category **Overcoming** is related to the process of coping with challenges within software project environments.
- **Promoting inclusion and equity** category addresses seven suggestions to improve the software industry in relation to women presence and equal treatment compatible with the competence of women in relation to men.
- The **Professional Development** category includes suggestions that encourage the dissemination of courses and mentoring aimed at better women training for the software industry.
- The category was called **Building networks and support** and includes suggestions that refer to creating connections and relationships to improve relationships, resource availability, opportunities, and positive results in the work environments of software projects for companies.

- The category **Promoting inclusion and Equity**, being the most cited, addresses seven suggestions to improve the software industry in relation to female presence and equal treatment compatible with women's competence in relation to men. In this category, we have for example the suggestion: *Inclusive organizational culture* with 12 mentions and can be explained as (A38): "Companies need to do more to create opportunities for individuals of diverse gender identities to advance to leadership positions and create a culture of inclusion that values diversity."



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Highlights

- The results reveal that women in software engineering face a **lack of confidence**, **fear of judgment**, and a **constant need to prove their competence**.
- They are often assigned to tasks based on **gender stereotypes**, such as documentation, while men take on technical roles.
- The results also reveal that a culture of **toxic masculinity** and **verbal harassment** discourage their participation..



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Highlights

- The study shows that gender inequality in software engineering causes **demotivation** and **burnout**, leading many women to leave the field early. This dropout reinforces the perception that programming is a predominantly male field, hindering inclusion and diversity in the sector.
- **Recognition of skills**, **promotion of gender awareness among colleagues** and **implementation of programs** are aspects listed in the study that contribute to a more inclusive environment, encouraging innovation and allowing greater professional advancement for women in the sector.



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The challenges faced by women in software engineering (SE), as identified in this GL study, reveal a concerning reality. Previous research confirms significant challenges to women's participation and advancement in the industry. They often deal with a lack of confidence and fear of judgment from men colleagues, which negatively impacts their performance and self-esteem. The constant need to prove competence is also highlighted as a major obstacle.

Task allocation based on gender stereotypes is common, with women often assigned non-technical roles, such as documentation, while men are given programming tasks [5].

This limits women's technical development and reinforces the stereotype that programming is men-dominated. Toxic masculinity and verbal harassment also discourage their participation. Increasing the proportion of women on teams reduces gender bias and promotes a more inclusive environment [27]. These challenges cause demotivation, exhaustion, and may lead many women to leave the tech industry early, exacerbating the underrepresentation of women in SE [5].

Several strategies have been proposed to address these challenges. The most common is recognizing women's skills and raising awareness about gender among men colleagues [4]. Mentoring within companies, maintaining confidence in one's abilities, and adopting gender equality policies are essential to improving recognition and promoting career advancement into leadership roles. Studies suggest that diverse teams lead to greater innovation and more effective solutions [28]

- Both studies identify **similar challenges**.
- The grey literature study highlighted **parental responsibilities, lack of family support**. Canedo et al. emphasized **cultural differences**.
- The GL study suggested **self-knowledge and confidence** and proposed **mental health and resilience**. Canedo et al. suggested **the use of language as a tool**.
- The GL study mentioned **support networks** as a solution.

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In their study, Canedo et al. [22] conducted qualitative interviews using grounded theory to identify three key categories: challenges, barriers, and suggestions. This approach aligns closely with the methodology used in our study, which also aimed to identify gender-related challenges faced by women in their work environments and to propose strategies for reducing gender disparities and mitigating these obstacles in women's professional lives.

In a comparative analysis, the findings of Canedo et al. [22] and our own are largely complementary, with some minor differences. Both studies identified common challenges, but the GL study highlights additional factors such as parental responsibilities and lack of family support. On the other hand, Canedo et al. [22] emphasized cultural differences as a significant challenge influencing gender disparity. Regarding the suggestions to address these challenges, the Canedo et al. [22] study identified use of language as a tool for improving gender relations, a suggestion not present in the GL study.

However, the GL study introduces other recommendations not found in Canedo et al. [22], including the importance of self-knowledge and confidence, mental health and wellbeing, resilience, and the development of networks and support systems.

Highlights - Advantages of Grey Literature (GL) Research

- Practical perspective on industry challenges.
- Faster updating than academic publications.
- Identification of emerging issues.
- Inclusion of diverse voices in the debate.
- Greater understanding of gender inequalities.

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The GL study offers significant advantages by providing real-time, practitioner-driven insights that reflect current challenges and trends in the software engineering industry. Unlike traditional academic literature, which may have longer publication cycles, grey literature captures immediate, evolving issues and solutions shared by professionals actively engaged in the field. Additionally, the open and accessible nature of GL allows for a broader range of voices to be heard, including those from individuals who may not contribute to peer-reviewed studies. This inclusive approach enhances the research by offering diverse perspectives and a more comprehensive understanding of gender inequality in the industry.

This study serves as a call to action to transcend complacency and confront inequities rooted in the software industry. By adopting inclusive and respectful practices, we can create a brighter, more equitable future where all professionals can thrive and contribute to the advancement of technology. The strategies suggested here, based on reports from the GL, offer a starting point for companies and organizations that want to promote gender equality and diversity in their workplaces.

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- We intend to continue supporting research and practical solutions aimed at **gender equality and the improvement of women's conditions in the workplace**, particularly in areas related to Information and Communication Technologies.
- For the next phase of this study, we intend to consider data from other sources of GL, such as **InfoQ** and **Stack Exchange**.



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Questions and Remarks?

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Thank you!



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